SDG Blockchain Accelerator Cohort 2
Deliverables Roadmap

Accelerator Duration: 29 September - 15 December 2025
Challenge Owner (CO): UNDP INDIA
Solution Maker (SM): PLASTIKS.IO (NOZAMA TECH LTD.)

1. Challenge Definition

1.1 Refined Challenge Statement

Describe the real-world development challenge your solution addresses.

*The challenge addressed is the **lack of transparent, inclusive, and verifiable systems for plastic waste recovery and recycling**, particularly within informal waste picker ecosystems. Waste pickers play a foundational role in plastic recovery but remain **invisible, unverified, and excluded from digital and financial benefits**, while brands and governments lack trusted mechanisms to verify real recycling outcomes and prevent fraud.*

1.2 Local Context

(Why does this challenge matter? Describe the setting and key beneficiaries. Explain why this challenge is important in your specific country or regional context. Describe the affected communities or sectors, any existing efforts to solve the issue, and the roles of key stakeholders involved.)

*This challenge is especially critical in **India**, where plastic waste recovery relies heavily on **informal waste pickers** operating without formal recognition, digital access, or social protection. Key issues include:*

- *No end-to-end traceability from waste picker to recycler*
- *Limited or no access to smartphones or digital wallets*
- *Absence of direct benefit distribution to workers*
- *Difficulty for brands and authorities to verify genuine recycling*

*Existing efforts involve NGOs, aggregators, recyclers, and urban local bodies (ULBs), but these are **fragmented and largely offline**, creating data gaps and exposure to fraud. The*

*most affected beneficiaries are **informal waste pickers and their communities**, while brands, recyclers, and regulators struggle with compliance and reporting.*

1.3 Alignment with UNDP Country Office and National Development Priorities

Explain how the proposed solution aligns with the strategic objectives of the UNDP Country Office and national development frameworks. Does it support existing programs or government priorities (e.g., digital transformation, climate resilience, youth employment, financial inclusion)? Highlight any synergies with UNDP initiatives, government strategies, or donor-funded efforts.

*The solution aligns with the **UNDP India** priorities by supporting:*

- ***Digital inclusion** of informal workers (QR-based IDs, no smartphones required)*
- ***Climate action and plastic pollution mitigation** through verified recycling*
- ***Formalisation of informal sectors** and welfare visibility*
- ***EPR, CSR, and ESG compliance** via tamper-proof impact data*

*The pilot directly complements national and UNDP-supported agendas on **digital public infrastructure, circular economy, climate resilience, and inclusive development***

Specifically, The proposed solution aligns closely with the strategic objectives of the UNDP Country Office and India's national development priorities by addressing intersecting challenges related to digital inclusion, environmental sustainability, and inclusive economic development.

The pilot supports **digital inclusion of informal workers** by enabling participation through QR-coded digital identities linked to custodial wallets, without requiring smartphones, internet access, or advanced digital skills. This approach is aligned with national and UNDP-supported efforts to expand digital public infrastructure in a manner that is inclusive and accessible to underserved populations.

The solution advances **climate action and plastic pollution mitigation** by enabling verified tracking of plastic recovery and recycling. By generating accurate, auditable data on material flows, the pilot strengthens accountability within waste management systems and supports climate mitigation efforts linked to circular economy practices and reduced plastic leakage into the environment.

The pilot also contributes to the **formalisation of informal sectors and improved welfare visibility**. By documenting waste pickers' economic activities and contributions, the system

creates pathways for recognition, social protection, and integration into formal policy and regulatory frameworks, supporting inclusive growth objectives.

In addition, the solution strengthens **EPR, CSR, and ESG compliance** by providing tamper-proof, verifiable impact data for governments, brands, and development partners. This directly supports national regulatory priorities related to Extended Producer Responsibility and enhances the credibility of sustainability reporting.

Overall, the pilot complements UNDP-supported and national agendas on **digital public infrastructure, circular economy development, climate resilience, and inclusive development**, and demonstrates how emerging technologies can be applied pragmatically to strengthen public systems, inform policy, and enable scalable impact.

1.4 Relevance to UNDP Programmes and Resource Mapping

(Is the pilot building on an existing initiative, or is it a new standalone project? If the project builds on an existing initiative, provide further details. Include scope, scale, timeline and available co-financing or in-kind contributions.

*The pilot is a **new implementation** that builds on Plastiks' existing blockchain infrastructure while aligning with UNDP pilot frameworks used previously (e.g. UNDP Armenia reference model).*

Scope & Scale

- *End-to-end plastic traceability from waste picker to recycler*
- *Multi-actor participation (pickers, aggregators, recyclers, NGOs)*

Timeline

- *Pilot-phase deployment with weekly coordination and demo milestones*

Resources & Co-financing

- *Platform deployment by Plastiks*
- *NGO facilitation and audits*
- *Targeted donor and blended-finance mobilisation (e.g. Fondation Valéry model)*

1.5 Key Stakeholders & Potential Partners

(Please list all the partners involved in this project. List all relevant parties who will interact with, benefit from, or influence the solution (this may include government agencies, NGOs, community members, or tech partners.))

- *Informal waste pickers*
- *Aggregators and large aggregators*
- *Plastic recyclers*
- *NGO partners (profiling, audits, facilitation)*
- *Urban Local Bodies (ULBs)*
- *Brands subject to EPR/CSR obligations*
- *UNDP India (oversight and validation)*
- *Plastiks (technology and platform provider)*

1.6 Expected Impact

(Outline intended outcomes from the CO's perspective. What would a successful pilot enable - policy change, improved services, empowerment, systems improvement, or scaling innovation?)

From the Country Office perspective, a successful pilot would enable:

- ***Formalisation and recognition** of waste pickers*
- ***Improved income and social protection** through direct credit benefits*
- ***Fraud-resistant, auditable recycling data***
- ***Policy-ready evidence** for EPR implementation*
- *A scalable, replicable model for national and regional deployment*

Specifically, from the Country Office perspective, a successful pilot would demonstrate how digital public infrastructure can be applied to address structural gaps in plastic waste management while advancing social inclusion and environmental accountability.

*The pilot would enable the **formalisation and recognition of informal waste pickers** by providing them with verifiable digital identities, transaction histories, and documented contributions to recycling value chains. This recognition supports improved visibility of informal workers within public systems and creates pathways for their inclusion in welfare, social protection, and policy frameworks.*

*Through automated and transparent credit distribution, the pilot would lead to **improved income stability and access to social protection benefits** for waste pickers. By linking verified plastic recovery to direct benefits, the system empowers workers economically while reducing dependency on opaque, cash-based transactions.*

*The pilot would generate **fraud-resistant, auditable recycling data**, addressing a critical gap faced by governments, development partners, and brands. Immutable records of plastic recovery and recycling strengthen trust in reported outcomes and significantly reduce risks of double counting, misreporting, or unverifiable claims.*

*By producing reliable, granular, and verifiable data, the pilot would provide **policy-ready evidence to support Extended Producer Responsibility (EPR) implementation**. This includes data that can inform regulatory design, compliance monitoring, and performance assessment at municipal, state, or national levels.*

*Finally, the pilot would validate a **scalable and replicable model** that can be adapted for broader national deployment and regional replication. Its modular design allows integration with existing waste management systems and UNDP-supported programmes, positioning the solution as a foundation for scaling innovation in circular economy, climate action, and inclusive development.*

1.7 Target SDGs and Indicators

(List up to three specific SDG indicators your challenge and proposed solution will directly contribute to. Focus on indicators you will actively address, not indirect ones.)

The pilot directly contributes to the following Sustainable Development Goal (SDG) indicators through measurable, on-the-ground activities and verified data generation:

- **SDG 12.5.1 – National recycling rate, tons of material recycled**
  The solution directly increases and accurately measures the volume of plastic waste that is recovered and recycled by enabling end-to-end traceability from collection to final recycling. Each kilogram of plastic handled within the pilot is digitally recorded, verified, and auditable, generating reliable data on actual recycling outcomes. This supports improved national recycling metrics and strengthens evidence-based reporting on waste reduction and circular economy performance.

- **SDG 8.3.1 – Proportion of informal employment formalised**
  By providing informal waste pickers with QR-coded digital identities, transaction records, and verifiable participation in formal recycling value chains, the pilot contributes to the gradual formalisation of informal work. Waste pickers gain documented proof of economic activity, access to structured benefit mechanisms,

and recognition within regulated systems, supporting pathways toward inclusive economic growth and decent work.

- **SDG 13.2.1 – Integration of climate measures into policies and reporting**
  The pilot generates tamper-proof, verifiable data on plastic recovery and recycling, which can be used by governments, UNDP, and partners to support climate-related reporting and policy integration. By linking material recovery data to auditable digital records, the solution strengthens the credibility of climate mitigation actions within waste management, EPR frameworks, and sustainability reporting processes.

2. User Mapping & Testing Strategy

2.1 Primary User Persona

Describe the main user or beneficiary of your solution.

The primary user is an **informal waste picker** working at the base of the plastic recycling value chain. This individual typically operates without access to a smartphone, reliable internet connectivity, or formal identification linked to digital systems. Their income is largely dependent on daily transactions with aggregators, often with limited transparency around pricing, volumes, or downstream value.

The waste picker's work is essential to plastic recovery but remains largely invisible in formal systems. They lack official recognition, have limited access to social protection mechanisms, and are excluded from most digital finance, data, and reporting frameworks. Despite contributing significantly to environmental outcomes, they do not benefit from verified recycling claims, credit mechanisms, or policy-level interventions.

The proposed solution is designed around this user's constraints and realities, enabling participation through QR-coded identities, simple verification processes, and indirect access to digital benefits. By addressing barriers related to technology, trust, and recognition, the system seeks to improve income stability, access to community benefits, and integration into more formal and transparent recycling ecosystems.

2.2 User Story

Use this format: "As a [user], I want to [goal], so that [value]."

As an informal waste picker, I want the plastic I collect to be digitally recorded and verified at the point of handover, so that my work is formally recognised, my contribution to recycling is visible, and I receive fair and transparent benefits. I want this system to work without requiring a smartphone or technical skills, and to allow me to access income, protective equipment, health services, or community benefits linked to my recycling activities. Through this, I am able to improve my livelihood, gain social recognition, and participate more securely in the formal recycling value chain.

2.3 Testing Strategy

How and when will user testing be conducted?

User testing is conducted during live pilot deployment to ensure the solution functions effectively under real operational conditions. Testing focuses on validating the system across all stages of the recycling value chain, using actual plastic recovery transactions rather than simulated data.

Aggregators test the system by recording real waste picker transactions, including QR-ID scanning, material classification, weight entry, and transaction confirmation. This validates the usability of the onboarding process, the accuracy of data capture, and the reliability of transaction logging.

QR-coded IDs are tested in field conditions to ensure consistent scanning, correct linkage to custodial wallets, and accurate credit allocation to waste pickers at the point of material handover. This confirms that the system operates effectively without requiring smartphones or digital literacy from end users.

Recyclers participate by verifying incoming plastic batches and uploading proof-of-recycling documentation. This testing step validates the final stage of the traceability loop, ensures correct batch reconciliation, and confirms that credit generation and distribution are triggered only after verified recycling.

Testing is continuous throughout the pilot and is reviewed during weekly coordination checkpoints with UNDP and key stakeholders. Findings from each testing cycle are documented and used to refine workflows, user guidance, and system configurations, enabling iterative improvement and risk mitigation during the pilot phase.

2.4 Feedback Loop Design

How will feedback from users and stakeholders be gathered and integrated?

Feedback is collected continuously throughout the pilot through multiple structured channels to ensure both operational effectiveness and stakeholder accountability. NGO facilitators

play a central role by working directly with waste pickers and aggregators, gathering qualitative feedback on onboarding, transaction recording, benefit distribution, and inclusion challenges. Periodic audits conducted with NGO and local partners help identify data inconsistencies, usability issues, and potential risks related to fraud or exclusion.

Operational feedback is also collected from aggregators and recyclers, focusing on transaction workflows, data capture processes, proof-of-recycling submissions, and system usability. This feedback helps refine operational protocols, training materials, and user interfaces to ensure the system integrates smoothly into existing recycling practices.

In parallel, UNDP and key stakeholders review blockchain records, dashboards, and pilot performance reports to assess data integrity, transparency, and alignment with pilot objectives. Insights from these reviews are systematically documented and translated into platform improvements, updated smart contract parameters where required, and revised operational guidelines. This structured feedback loop ensures continuous learning, accountability, and iterative improvement throughout the pilot lifecycle.

Feedback Flow:

Waste Pickers
→ Feedback collected by NGO facilitators (onboarding, usability, benefit access)
→ NGO reports and periodic audits

Aggregators & Recyclers
→ Operational feedback (transactions, data entry, verification processes)
→ Technical and process review

NGOs, UNDP & Key Stakeholders
→ Review of blockchain records, dashboards, and audit findings
→ Identification of issues, risks, and improvement opportunities

Plastiks Platform & Operations Team
→ Platform adjustments (workflows, rules, interfaces)
→ Updates to operational protocols and training materials

Revised System
→ Improved user experience, data quality, and transparency
→ Continuous feedback cycle during pilot implementation

3. Solution Overview

3.1 Solution Summary

Briefly describe your proposed blockchain-based solution.

The proposed solution is a **blockchain-based** plastic waste traceability and credit distribution system designed to make informal plastic recovery visible, verifiable, and inclusive. The system digitally records the full lifecycle of plastic waste—from collection by informal waste pickers through aggregation and final recycling—creating a transparent and auditable chain of custody for each material batch.

Using Q**R-coded digital identitie**s linked to custodial wallets, the solution enables waste pickers to participate in a digital system **without the need for smartphones, internet access, or technical skills.** At each stage of the value chain, material quantities and transactions are securely recorded, and verified credits are automatically generated per kilogram of plastic recovered and recycled. These credits are distributed directly to all contributing actors according to predefined rules, ensuring fair benefit-sharing and reducing reliance on manual processes.

By combining **immutable blockchain records with recycler verification**, the system prevents fraud, improves data reliability, and produces trusted datasets that can support Extended Producer Responsibility (EPR), CSR, and ESG reporting. The solution strengthens social inclusion, enhances transparency across the recycling ecosystem, and provides a scalable digital public infrastructure for circular economy initiatives and policy implementation.

3.2 Core Features and Functionalities

What are the most important features of your solution?

- **QR-Coded Digital IDs Linked to Custodial Wallets**
  Each waste picker is onboarded through NGO partners and issued a unique QR-coded ID. This QR code is linked to a custodial blockchain wallet, allowing waste pickers to participate in the system without requiring a smartphone, internet access, or digital literacy. The QR code functions as a persistent digital identity across all transactions, ensuring inclusion, preventing duplicate or ghost identities, and enabling accurate attribution of collected material and benefits.

- **End-to-End Material Tracking**
  Plastic material is tracked from waste picker to aggregator, from aggregator to large aggregator, and finally to the recycler. Every transaction records key data such as actor identity, material type, weight, and timestamp. This information is logged digitally at each handover, creating a continuous and verifiable chain of custody for every plastic batch and ensuring full transparency across the recycling value chain.

- **Automated Credit Generation and Allocation**
  Credits are automatically generated per kilogram of verified plastic at each value-adding stage of the chain. Smart contracts define and enforce how credits are allocated among waste pickers, aggregators, and recyclers. This automation ensures transparent, rule-based distribution of benefits, reduces disputes, removes the need

for manual reconciliation, and guarantees that informal workers receive direct and fair value for their contribution.

- **Recycler Proof-of-Recycling Uploads**
  Recyclers are required to confirm receipt and processing of plastic batches by uploading proof of recycling, such as documentation or photographic evidence. This step validates the final stage of the material lifecycle and triggers the release of final credits. It ensures that credits are issued only for plastic that has been genuinely recycled, strengthening trust for brands, regulators, and development partners.

- **Tamper-Proof, Auditable Records**
  All transactions and material movements are recorded on an immutable blockchain ledger. Each plastic batch has a permanent digital identity and a complete transaction history that cannot be altered retroactively. This design prevents fraud and duplication, enables independent audits by NGOs and public institutions, and provides reliable, verifiable data for EPR, CSR, and ESG reporting and policy decision-making.

3.3 Tech Stack and Tools

Which technologies and platforms will you use?

The solution will use a **hybrid architecture** combining a user-friendly digital platform with blockchain-based integrity and certification, ensuring it is practical for field use while meeting institutional requirements for transparency and auditability.

- **Web-based platform (user interface + admin dashboards)**
  A browser-accessible platform for recyclers, aggregators, and partner organizations to submit recovery data, manage verification steps, and view dashboards and reports.

- **Blockchain layer (traceability + immutable audit trail)**
  A blockchain network to record verified recovery events and certification outcomes, ensuring tamper-proof records, preventing double counting, and enabling transparent auditing.

- **Smart contracts / on-chain logic (where required)**
  Used to automate key blockchain interactions such as certification status recording and rules for credit issuance or proof-of-recovery statements.

- **Off-chain database and storage (operational data + documents)**
  A secure off-chain database for operational records (user accounts, workflow status, metadata) and document storage for supporting evidence, with hashes or references anchored on-chain where appropriate.

- **Data analytics and reporting tools**
  Built-in reporting dashboards and exportable reports for UNDP, government partners, and sponsor companies to support monitoring, compliance, and donor/corporate reporting.

- **User access management and security**
  Role-based access control, authentication, and audit logs to ensure appropriate permissions and accountability across recyclers, aggregators, and institutional stakeholders.

This stack ensures a balance between **field usability**, **scalability**, and **institutional-grade transparency**, while keeping the solution flexible for future integration into EPR systems and national reporting frameworks.

3.4 Blockchain Functionality

What blockchain-specific interactions are included (e.g., token minting, smart contracts, on-chain verification)?

The blockchain component is used as a **trust and integrity layer** to ensure that plastic recovery data and certification outcomes are **tamper-proof, auditable, and credible** for government, UNDP, donors, and private sector partners. Key blockchain-specific interactions include:

- **On-chain recording of verified recovery events**
  Once plastic recovery data is submitted and validated through the verification workflow, a corresponding record (or cryptographic proof) is written to the blockchain. This creates an immutable audit trail that prevents manipulation and supports transparent reporting.

- **On-chain certification / proof-of-recovery status**
  The platform records certification outcomes on-chain (e.g., “verified,” “certified,” “rejected”), ensuring that verification decisions are traceable and can be independently audited by authorized stakeholders.

- **Smart contract–based verification logic (as applicable)**
  Smart contracts can be used to enforce standardized rules for recording verification results, timestamps, and actor roles (e.g., recycler → aggregator → verifier), ensuring consistent application of certification criteria and reducing disputes.

- **Prevention of double counting and fraud risk reduction**
  By anchoring verified recovery records on-chain, the solution reduces the risk of duplicated claims and supports integrity checks across actors and locations—critical

for EPR compliance and credible impact finance.

- **Plastic Credit enablement (issuance linkage)**
  Certified recovery records can be linked to the generation/issuance of **Plastic Credits** (or equivalent environmental units). Where credit issuance is implemented, blockchain records provide the provenance and traceability required to demonstrate that credits correspond to verified recovery.

- **Auditability for institutional and donor reporting**
  Blockchain provides an independent, time-stamped record that enables UNDP and partners to demonstrate transparency, strengthen accountability, and support results-based financing or donor assurance requirements.

These blockchain interactions ensure the solution delivers **verifiable traceability and certification**, enabling credible monetization mechanisms and long-term readiness for EPR implementation.

4. Weekly Workplan

Briefly describe your team's focus during each stage of the program. (You may use the workplan template from your hackathon roadmap.)

Week 1 (1–7 February 2026)
Project Kickoff and Internal Alignment
The pilot begins with a formal project kickoff. The project team is introduced, roles and responsibilities are confirmed, and budget approval is finalised. Initial coordination meetings are held to align objectives, scope, and governance arrangements with UNDP India and core partners.

Week 2 (8–14 February 2026)
Working Meetings and Stakeholder Outreach
The team conducts internal and external working meetings and initiates structured stakeholder outreach. Engagement focuses on NGOs, aggregators, recyclers, and relevant ecosystem actors to prepare them for pilot participation.

Week 3 (15–21 February 2026)
Stakeholder Presentations and Commitment Collection
The solution and pilot framework are formally presented to identified stakeholders. This phase focuses on collecting confirmations, commitments, and participation agreements required for implementation.

Week 4 (22–28 February 2026)
Accreditation and Documentation
Accreditation processes are initiated and completed for participating entities. Required

documentation, compliance checks, and validation steps are finalised to ensure operational and regulatory readiness.

Week 5 (1–7 March 2026)
Stakeholder Onboarding and System Preparation
Approved stakeholders are onboarded into the platform, including NGOs, aggregators, and recyclers. System configurations are completed in preparation for verification and tokenization workflows.

Week 6 (8–14 March 2026)
Verification and Tokenization
Initial plastic recovery activities are verified, and blockchain-based tokenization and credit generation processes are activated. Real pilot transactions are tested under live conditions.

Week 7 (15–21 March 2026)
Plastic Credit Validation and Readiness for Allocation or Sale
Generated plastic credits are reviewed and validated. The team prepares credits for allocation or sale, ensuring data integrity, traceability, and alignment with pilot objectives.

Week 8 (22–28 March 2026)
Training and Onboarding Centre Setup
Training sessions are delivered to operational partners, and onboarding centres are set up to support waste pickers and stakeholders. Final readiness checks are conducted to ensure pilot continuity and potential scale-up.

5. Final Deliverables

5.1 Solution Demo (PoC)

A working or visual prototype that demonstrates the core idea. (To be scheduled and presented virtually to CO team and UNDP/Partner FPs in a weekly coordination meeting)

5.2 Testing & Feedback Documentation

Summary of user testing activities, challenges, and feedback.

5.3 Feedback Integration Summary

Description of what changes were made based on testing and feedback.

5.4 Pilot-Ready Solution

A stable and functional version of the solution, ready for pilot deployment.

5.5 Confirmed Pilot Partnership(s)

Documented confirmation of at least one partner committed to implementation.

5.6 Stakeholder Alignment Matrix

Defined roles, responsibilities, and coordination plan with key partners.

Stakeholder / Partner	Role in the Pilot	Key Responsibilities	Level of Involvement (Lead / Support / Advisory)	Engagement Modality (e.g. MOU, Contract)	Focal Point / Contact Person

5.7 Budget Proposal (Pilot)

Provide a budget breakdown for pilot implementation. Include estimated cost, unit, quantity, and funding source.

Budget Line Item	Unit / Description	Quantity	Unit Cost (USD)	Total Cost (USD)	Funding Source (Confirmed / Pending)
Project Management & Coordination	Pilot project management and reporting	1 pilot	5,000	5,000	Pending (UNDP / Donor)

Platform Deployment & Configuration	Deployment of blockchain traceability platform	1 pilot	6,000	6,000	Pending (UNDP / Donor)
Accreditation of Recycling & Recovery Entities	Accreditation of aggregators, recyclers, and recovery entities	3 entities	1,200	3,600	Pending (Donor / Co-financing)
Licensing, Verification & Audits	Annual licensing, verification, and audits	1 package	3,000	3,000	Pending (UNDP / Donor)
NGO Facilitation & Waste Picker Onboarding	Profiling, QR-ID issuance, onboarding support	1 package	3,500	3,500	Pending (UNDP / Donor)
Monitoring, Reporting & Compliance Data	Impact reporting, dashboards, UNDP reviews	1 package	1,900	1,900	Pending (UNDP / Donor)
Contingency & Operational Buffer	Field operations and unforeseen costs	1	2,000	2,000	Pending
Training & Onboarding Centres	Training sessions and onboarding centre setup	1 package	0	0	In-kind (Plastiks)
TOTAL ESTIMATED COST				**25,000**	

5.8 Impact Measurement Framework

5.8.1 Contribution to relevant SDGs and indicators

The pilot contributes directly to selected Sustainable Development Goals (SDGs) by embedding measurable outcomes into the design and operation of the blockchain-based traceability and credit distribution system. Contributions are grounded in verifiable data generated through real pilot activities rather than indirect or assumed impacts.

SDG 12: Responsible Consumption and Production

Indicator 12.5.1 – National recycling rate, tons of material recycled

The pilot directly supports this indicator by enabling accurate measurement of plastic waste that is collected, aggregated, and recycled. Each kilogram of plastic handled within the pilot is digitally recorded at every stage of the value chain—from waste picker to recycler—and verified through recycler proof-of-recycling uploads. The system generates auditable, tamper-proof data on actual recycling volumes, reducing risks of double counting or misreporting. This improves the quality and reliability of recycling statistics and supports evidence-based monitoring of waste reduction and circular economy performance.

SDG 8: Decent Work and Economic Growth

Indicator 8.3.1 – Proportion of informal employment formalised

The pilot contributes to the formalisation of informal waste pickers by providing them with QR-coded digital identities, documented transaction histories, and verified participation in structured recycling systems. Waste pickers' economic activities are recorded and recognised within a transparent digital framework, creating visibility that is typically absent in informal work. This enables pathways toward formal recognition, access to benefit mechanisms, and inclusion in policy and welfare discussions, directly supporting inclusive economic growth and improved working conditions.

SDG 13: Climate Action

Indicator 13.2.1 – Integration of climate measures into policies, strategies, and reporting

By producing verified and auditable data on plastic recovery and recycling, the pilot strengthens the integration of climate mitigation measures into reporting and policy processes. The traceability system links material recovery outcomes to immutable records, enabling governments, UNDP, and partners to credibly incorporate waste-sector climate actions into sustainability reporting, EPR frameworks, and climate-related strategies. This enhances transparency and accountability in climate action related to plastic pollution and resource efficiency.

Cross-Cutting Impact

Beyond individual indicators, the pilot demonstrates how digital public infrastructure can support multiple SDGs simultaneously by combining environmental outcomes, social inclusion, and data integrity. The system generates high-quality datasets that can be reused for policy design, regulatory enforcement, development programming, and scaling initiatives, reinforcing UNDP's integrated approach to sustainable development.

5.8.2 Key Performance Indicators (KPIs)

KPI	*Measurement Method*	*Target / Threshold*	*Notes*
Number of waste pickers onboarded	*NGO profiling records and QR-coded ID issuance logged on the platform*	*Pilot cohort onboarded as per agreed scope*	*Measures inclusion of informal workers without smartphone dependency*
Percentage of waste pickers using QR-coded IDs	*Ratio of QR-linked transactions to total transactions*	*≥ 90% of pilot transactions*	*Indicates adoption and usability of QR-based identity system*
Total plastic collected and recorded (kg)	*Aggregator transaction logs recorded on blockchain*	*Verified baseline volume achieved during pilot*	*Direct input for recycling and impact reporting*
Percentage of plastic batches fully traceable end-to-end	*Blockchain records from waste picker to recycler*	*100% of pilot batches*	*Confirms complete chain-of-custody tracking*
Number of verified recycling transactions	*Recycler proof-of-recycling uploads linked to batches*	*All batches reaching recycler stage*	*Ensures credits issued only for recycled plastic*

Credits generated per kilogram of plastic	*Smart contract and blockchain credit records*	*Credits generated for 100% of verified kg*	*Validates automated credit creation*
Percentage of credits allocated to waste pickers	*Custodial wallet credit distribution logs*	*Predefined share executed automatically*	*Confirms direct benefit delivery to workers*
Number of credit redemptions by waste pickers	*Redemption records (cash-out, PPE, services)*	*At least one redemption instance during pilot*	*Demonstrates access to benefits without phone usage*
Incidents of duplicate or fraudulent records	*Audit findings and blockchain integrity checks*	*Zero confirmed incidents*	*Core integrity and transparency indicator*
Availability of auditable compliance data	*Blockchain dashboards and exportable reports*	*100% availability for pilot duration*	*Supports EPR, CSR, and ESG reporting*
Stakeholder participation rate	*Active participation of NGOs, aggregators, recyclers*	*All committed stakeholders active*	*Measures ecosystem engagement*
Completion of periodic audits	*NGO and facilitator audit reports*	*Audits completed as scheduled*	*Supports UNDP oversight and data credibility*

5.8.3 Monitoring & Reporting Plan

Frequency:
Monitoring is conducted on a **continuous basis** through real-time transaction logging on the blockchain. Operational reviews are conducted **weekly** during pilot implementation, aligned with coordination checkpoints. **Periodic audits** are carried out at defined milestones during the pilot to validate data integrity, inclusion, and compliance.

Responsible Parties:

- **Plastiks team:** Responsible for platform operation, system monitoring, data integrity checks, and generation of dashboards and reports.

- **NGO partners:** Responsible for waste picker profiling, field-level verification, facilitation, and periodic audits.

- **Aggregators and recyclers:** Responsible for accurate transaction recording and submission of proof-of-recycling documentation.

- **UNDP India:** Provides oversight, reviews reports and dashboards, validates pilot progress, and ensures alignment with programme objectives.

Format:
Monitoring outputs are provided through:

- Digital dashboards displaying transaction volumes, traceability status, and credit distribution

- Periodic summary reports (weekly or milestone-based) covering operational performance, risks, and corrective actions

- Audit summaries documenting verification outcomes and data quality checks

All reports are structured to support UNDP programme monitoring, SDG reporting, and EPR/ESG compliance needs.

Tools:

- Blockchain-based traceability platform with immutable transaction records

- QR-coded identity system linked to custodial wallets

- Smart contracts for automated credit generation and allocation

- Exportable data reports and dashboards for UNDP and stakeholder review

- Audit tools and verification protocols implemented by NGO partners

5.8.4 Risks Assessment & Mitigation Plan

Risks	Mitigation
Low adoption by waste pickers due to lack of digital literacy or trust	Use QR-coded IDs with no smartphone requirement; onboarding and facilitation led by trusted NGO partners; clear communication of benefits and redemption options
Inaccurate or incomplete data entry by aggregators	Training provided to aggregators; simple transaction workflows; real-time validation checks; periodic NGO-led audits
Fraud, duplication, or ghost worker records	Unique QR-coded identities linked to blockchain wallets; immutable blockchain records; photo and ID verification; periodic audits
Failure to verify final recycling outcomes	Mandatory recycler proof-of-recycling uploads before final credit issuance; batch reconciliation at recycler level
Delays in stakeholder onboarding or accreditation	Early stakeholder outreach and commitment collection; clear accreditation requirements; phased onboarding approach
Resistance from recyclers or aggregators due to added reporting requirements	Streamlined digital processes; alignment with EPR and ESG reporting needs; demonstration of compliance and market benefits
Data misuse or lack of trust in reported outcomes	Tamper-proof blockchain ledger; auditable transaction history; UNDP and NGO oversight
Operational disruptions in field implementation	Contingency planning; operational buffer in budget; flexible timelines within pilot period
Sustainability of the model beyond the pilot	Design aligned with EPR, CSR, and ESG frameworks; preparation for donor and

blended-finance resource mobilisation; scalable system architecture

5.9 Post-Acceleration Roadmap & Sustainability Plan

5.9.1 Phased Implementation Timeline / Milestones

Short Term (0–6 months)

Pilot Implementation and Validation

- Formal project kickoff, governance setup, and coordination with UNDP India and partners
- Stakeholder onboarding, including NGOs, aggregators, and recyclers
- Profiling and onboarding of waste pickers using QR-coded digital IDs
- Deployment of the blockchain-based traceability and credit distribution platform
- Recording of real plastic recovery transactions from waste picker to recycler
- Verification of recycling outcomes through recycler proof uploads
- Initial credit generation, allocation, and benefit redemption testing
- Continuous monitoring, weekly reviews, and periodic audits
- Pilot evaluation and documentation of lessons learned

Key milestone: Functional, validated pilot demonstrating end-to-end traceability, inclusion of informal workers, and auditable recycling data.

Medium Term (6–18 months)

Scale-Up and System Integration

- Expansion of the solution to additional geographies, waste picker groups, and recycling partners
- Increased volumes of plastic recovery and recycling recorded on the platform

- Integration with Extended Producer Responsibility (EPR), CSR, and ESG reporting processes
- Strengthening of monitoring, dashboards, and reporting for government and UNDP use
- Engagement with additional donors, brands, and development partners for co-financing
- Refinement of operational protocols, audits, and governance mechanisms

Key milestone: Scaled deployment supporting policy-relevant data generation and broader ecosystem participation.

Long Term (18+ months)

Institutionalisation and Replication

- Integration of the system into national or sub-national waste management and EPR frameworks
- Institutional adoption by public authorities, regulators, and industry stakeholders
- Replication of the model across additional states or regions
- Use of verified data to inform policy design, regulatory enforcement, and climate reporting
- Establishment of a sustainable financing and operating model for long-term continuity

Key milestone: Institutionalised, scalable digital public infrastructure supporting circular economy, climate action, and inclusive development outcomes.

5.9.2 Budget for Phased Implementation (post-acceleration)

5.9.3 Financial Sustainability Model

How will the solution generate income post-pilot? Pricing model (e.g. subscriptions, licensing, transaction fees) and when it kicks in.

Overview

Post-pilot, the solution operates on a **volume-based usage pricing model** directly linked to verified material flows. Revenue is generated through a **base platform usage fee of USD 0.01 per kilogram verified (USD 10 per tonne)**, with **volume discounts applied at higher throughput levels** to incentivise scale and long-term adoption.

Commercial pricing applies **immediately after pilot completion and formal sign-off**, once the system enters operational use.

Revenue Model

Platform Usage Fee (Unified, Volume-Based)

Who pays:

- Corporates, producers, operators, EPR schemes, or institutions using the platform to verify material flows.

What it covers:

- Full access to the digital traceability and verification platform
- Batch submission and evidence validation
- Issuance of digital certificates / Plastic Credits
- Dashboards, audit logs, and reporting
- API access and system maintenance

Base Pricing:

- **USD 0.01 per kilogram verified**
- **Equivalent to USD 10 per tonne**

Volume Discounts:

- Tiered discounts apply automatically as verified volumes increase
- Discount levels are agreed contractually post-pilot and aligned with annual volume commitments
- Ensures declining unit cost at scale while maintaining platform sustainability

When it starts:

- **From the first commercial batch processed post-pilot**

Illustrative Revenue Examples (Before Discounts)

- **5,000 tonnes/year → USD 50,000/year**
- **25,000 tonnes/year → USD 250,000/year**
- **100,000 tonnes/year → USD 1,000,000/year**

(Actual payable amounts may be lower depending on applicable volume discount tiers.)

Financial Sustainability Logic

- **Pay-for-impact:** Fees are directly proportional to verified kilograms processed.
- **Scales efficiently:** Linear revenue growth with strong operating leverage.
- **Incentivises adoption:** Volume discounts reward growth and long-term commitments.
- **Low friction:** No fixed subscriptions or per-user fees simplify procurement.
- **Aligned incentives:** Platform economics grow with increased recovery and verification.

Assumptions

- Pilot confirms operational readiness and stakeholder alignment
- Commercial volumes ramp within 1–3 months post-pilot
- Volume thresholds and discount tiers are mutually agreed

5.9.4 Partnership and Co-Financing Commitments
Which partners will be supporting the project (technical, financial, institutional) and what would be their contributions?

5.9.5 Fundraising Plan

Plan for securing funding through grants, ecosystem funds, investors or other sources. Targeted sources and fundraising timeline.

5.11 Legal & Compliance Considerations

Summary of key legal, regulatory, or compliance considerations.
Overview

The solution is designed as a digital traceability and verification infrastructure, not as a regulated financial product. Legal and compliance considerations therefore focus on data protection, contractual enforceability, regulatory alignment (waste/EPR/ESG), and auditability, rather than financial regulation.

Data Protection & Privacy

The platform processes operational data, user accounts, and supporting evidence. It is designed to comply with GDPR and equivalent local data protection laws through data minimisation, encryption, role-based access controls, and clear data controller/processor agreements.

Waste, EPR & Environmental Regulation Alignment

The platform supports compliance with waste, EPR, and environmental reporting frameworks but does not replace statutory reporting or regulatory approval. All certificates or Plastic Credits function as digital evidence instruments.

Financial & Token-Related Considerations

Plastic Credits and digital certificates are not financial instruments, securities, or investment products. Platform fees are charged for verification services only, with clear legal disclaimers and separation from any third-party marketplaces.

Evidence Integrity, Auditability & Liability

Data originators remain responsible for submitted evidence. The platform provides traceability and audit trails but does not guarantee physical performance. Liability limits are defined contractually.

Cross-Border Operations

Multi-country deployments address cross-border data transfer safeguards and local regulatory reviews for waste and EPR frameworks.

Risk Summary

Overall legal risk is considered manageable through standard enterprise controls, GDPR-aligned architecture, jurisdiction-specific reviews, and clear contractual frameworks.

5.12 Visibility Assets / Communication Materials

1–2 slides, visuals, or storylines for use by UNDP or partners to promote your solution.

5.13 Pitch Deck

Concise and compelling presentation for stakeholders, funders, and partners. ((Please upload to your google drive folder)

5.14 Final Demo Presentation

An engaging walkthrough of the live solution and its implementation roadmap at the end of the program.

Deliverable Completion Table

Please indicate your intended completion week (or date) for each deliverable. You can revise this plan as the project evolves.

Deliverable	Target Completion (Week # or Date)
1. Challenge Definition (incl. local context, SDG contributions, partners, impact)	
2. User Mapping & Testing Strategy (personas, user stories, feedback loop)	
3. Solution Overview (core features, tech stack, blockchain functionality)	

5.1 Solution Demo (PoC)	
5.2 Testing & Feedback Documentation	
5.3 Feedback Integration Summary	
5.4 Pilot-Ready Solution	
5.5 Confirmed Pilot Partnership(s)	
5.6 Stakeholder Alignment Matrix	
5.7 Budget Proposal (Pilot)	
5.8 Impact Measurement Framework (KPIs, SDG indicators, risk plan)	
5.9 Sustainability Plan & Post-Acceleration Roadmap (implementation timeline, budget, partners, fundraising)	
5.11 Legal & Compliance Mapping	
5.12 Visibility Assets / Communication Materials	
5.13 Pitch Deck	